

Bacterial Genetics & Virulence - Factoid Sheet

Week 4 - Tuesday - Core Lecture 6 - current faculty deck + official Week 4 objectives

Deck mnemonic: Transformation = USB. Conjugation = cable. Transduction = virus email.

Three Routes Side by Side

Route	Vector	Contact?	DNase?	Cargo
Transformation	Naked DNA	No	Stops it	Small DNA/plasmid
Conjugation	Donor + pilus/adhesion	Yes	No effect	Large DNA/plasmids
Transduction	Bacteriophage	No	No effect	Small DNA/phage cargo

Transformation

Competent cell takes up free DNA → integrates by recombination or maintains plasmid. Naked DNA is exposed, so extracellular **DNase abolishes transformation**.

Conjugation

F plasmid
Classic Gram-negative donor-recipient transfer through direct contact/sex pilus.

Hfr
F factor integrated into chromosome; adjacent chromosomal genes can transfer.

Enterococcus twist: Gram-positive conjugation can be pheromone-driven; recipient peptide induces donor adhesins/clumping rather than a classic sex pilus.

Transduction

Type	Rule
Generalized	Random host DNA accidentally packaged
Specialized	Genes adjacent to prophage insertion site move after imprecise excision

Bacterial Genetics - Fast Recall

Resistance

Horizontal transfer spreads resistance faster than independent mutation in every lineage. Deck anchors: **mecA** in MRSA; **vanA** in VRE.

Lysogenic Conversion

Temperate phage integrates as a prophage; a phage gene changes bacterial phenotype, often by adding toxin production.

Toxin	Bacterium	Anchor
Diphtheria	C. diphtheriae	Diphtheria
Cholera	V. cholerae	Cholera
Shiga	STEC / E. coli O157:H7	Hemorrhagic colitis / HUS
Botulinum C/D	C. botulinum	Botulism
Erythrogenic	S. pyogenes	Scarlet fever

EHEC Clinical Logic

Prophage induction can increase Shiga toxin. Deck emphasizes hydration + HUS monitoring and avoidance of reflex antibiotics/antimotility therapy in suspected EHEC.

Historical Mechanism Anchors

Griffith/Avery: transformation and DNA as transforming principle.	Hershey-Chase: phage DNA enters bacteria.
Lederberg-Tatum: conjugation.	Zinder-Lederberg: transduction.

Must-Not-Miss Clues

Free DNA + DNase sensitive = transformation. Contact/pilus = conjugation. Phage = transduction. Prophage-encoded toxin = lysogenic conversion. Random transduced genes = generalized; adjacent-to-prophage genes = specialized.

Assigned ScholarRx

Bacterial Genetics & Virulence Factors: <https://exchange.scholarrx.com/brick/bacterial-genetics-and-virulence-factors>